HRIDIK HINGORANI

Curriculum Vitae

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Education

Aug 2023 – May 2027 (expected) **BS Computer Engineering, Minor in Mathematics**, *University of Illinois at Urbana-Champaign*

Coursework ECE 470 Robotics, ECE 486 Control Systems, ECE 484 Principles of Safe Autonomy, ECE 391 Operating Systems, ECE 494 Deep Learning for Computer Vision

Experience

May 2026 – Aug 2026 **AI Software Engineering Intern**, *Software Associates*, Remote

- Built a local-first voice assistant on embedded Linux and Apple Silicon: Whisper STT, LLM routing through Ollama with cloud fallback, Coqui TTS, a FastAPI service layer, and an Electron client. Ran MLX LoRA fine-tuning to adapt the local model's response behavior.
- Diagnosed and fixed audio-pipeline timing failures, device interface faults, and I2C/UART peripheral errors that were dropping utterances mid-stream, taking capture from intermittent to reliable end to end.
- Shipped a Python and Flask trading service on the Interactive Brokers REST API: a five-indicator weighted signal engine (RSI, MACD, Bollinger Bands, moving averages, ATR), automated stop-loss, take-profit and trailing-stop risk logic, and a live position and P&L dashboard.

May 2025 – Aug 2025 **AI Software Development Intern**, *HealthArk Insights*, Mumbai, India

- Built Python-based document-processing features including summarization, web scraping, and dynamic formatting tools tailored to client specifications.
- Integrated client feedback into production workflows by gathering requirements, validating features, and refining document-management functionality. Features shipped to production.

Projects

Jan 2024 – Aug 2025 **StethoSpy: Electronic Stethoscope**, *Project Lead, I-MADE UIUC*

Prototype electronic stethoscope for heart-sound acquisition and murmur screening, aimed at rural and low-infrastructure settings. Custom PCB captures heart sounds; an ADC/DAC pipeline processes the analog signal to flag cardiac irregularities for follow-up. Not clinically validated. Owned hardware design and signal processing end to end.

8-Channel EEG Acquisition Board, *Independent research*

8-channel, 2-layer EEG signal acquisition board in KiCad. Op-amp analog front end with low-pass filtering for microvolt-level biopotential acquisition; schematic capture through PCB layout designed for high signal integrity across all eight channels.

UR3 Vision-Guided Contour Drawing, *Python, ROS, OpenCV*

Closed-form analytic inverse kinematics for a 6-DOF arm, written from scratch, with product-of-exponentials forward kinematics evaluated through matrix exponentials. Drives a UR3 to draw an image: Canny edge detection and contour simplification mapped onto a 20 cm square drawing area, published as joint targets over ROS and held to 0.0005 rad per joint. A Pikachu test image came out as 142 strokes, drawn in 7 minutes.

Real-Time Task-Space Impedance Control: CRS Arm, *C, TMS320F28335 DSP*
Task-space PD impedance control on a three-link arm, mapped to joints through the Jacobian transpose and closed at 1 kHz on a timer interrupt. Reads joint encoder angles and outputs joint torque commands directly to the amplifiers; it does not send position setpoints to a servo controller. Per-waypoint stiffness modes give selective compliance, with per-joint Coulomb and viscous friction compensation and torque saturation at 5 N-m per joint. Contact tasks use a feedforward downward force commanded through the motor torque constant, with no force sensor and no closed-loop force feedback, validated afterward against a physical scale. Written in C on a TI TMS320F28335. Peg insertion, zig-zag tracking, and pushing an egg without breaking it.

We_Are_Kernel: RISC-V OS, *C, RISC-V*
A RISC-V kernel from scratch, roughly 6,000 lines of C: Sv39 paging, VirtIO, a FAT filesystem, fork/exec/exit, pipes, a shell with redirection, and preemptive multitasking.

Animatronic Grogu, *Python, C++, KiCad, ROS*
ECE 445 senior design capstone, currently in planning. A 22-DOF bipedal animatronic. Planned scope covers the software, the LLM voice pipeline (local LLM with Claude API, Whisper speech-to-text, Coqui text-to-speech), and the mechanical design, on a custom PCB for power.

Jupiter AI Voice Assistant, *Python, MLX, Qwen3, Whisper*
Fully local voice assistant for Apple Silicon, no cloud and no API keys. Whisper STT, a Qwen3 router and main model on MLX with speculative decoding and a persistent prompt cache, and Kokoro streaming TTS. Continuous VAD gives barge-in that halts generation mid-sentence. 0.3 to 0.6 s first token, warm.

Inverted Pendulum: Pole Placement and Observer, *MATLAB, Embedded*
Pole-placement state feedback with a Luenberger observer, built as two decoupled 2-state blocks and gain-scheduled across the upright and hanging linearizations. Friction parameters identified from logged step-response data. Measured 1.1-second settling time.

Teaching Experience

Jan 2026 – present | **Course Assistant, ECE 385 (FPGA Design)**, *University of Illinois at Urbana-Champaign*
Leads lab sections for 60+ students on SystemVerilog, FPGA timing closure, and hardware-software co-design. Debugs student implementations across RTL logic, timing constraints, and board-level signal integrity.

Jan 2025 – Dec 2025 | **Course Assistant, MATH 257 (Linear Algebra)**, *University of Illinois at Urbana-Champaign*
Taught Python-based linear algebra lab sections to approximately 200 students, supporting hands-on computational exercises.

Research

Ongoing | **Project Lead, Brainwaves Team**, *NeuroTechX UIUC*
Leads the Brainwaves project team, overseeing the club's podcast and social media output. EEG signal-processing and neural-network research through the club.

Ongoing | **Independent EEG Research**, *Self-directed*
Independently designed and built the 8-channel EEG acquisition board listed under Projects; separate from the club work.

Leadership

Jan 2025 – present | **Event Lead, then Senior Coordinator, National Team**, *Hindu YUVA*
Manages the UIUC chapter and the event team. Organizes large-scale Hindu cultural and community events at both chapter and national scope.

Jan 2024 – Aug 2025 | **Project Lead, StethoSpy**, *I-MADE UIUC*
Led the StethoSpy project (see Projects), owning hardware design and signal processing end to end.

Skills

Robotics & controls | Closed-form and analytic IK, product of exponentials, impedance and compliance control, LQR, PID, RK4 integration, motion planning (A*, RRT*), ZMP gait planning, ROS, URDF

Languages | C/C++, Python, SystemVerilog, RISC-V Assembly, MATLAB, Rust (application-level)

Embedded & hardware	Real-time Linux, bare-metal firmware, preemptive scheduling, I2C/UART/ADC/GPIO/PWM, SPI (loopback), ESP32, Raspberry Pi, FPGA, KiCad, Git

Off the Clock

Writing Poetry and literary fiction. The poems lean on Mumbai-specific references, Hinglish code-switching, and language kept precise. Posted, fairly regularly, on Instagram.